EVALUATION AND ADAPTATION TO FREQUENCY-TO-PLACE FACTORS IN COCHLEAR IMPLANT USERS

IHI – 3rd October 2026

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PhD Student

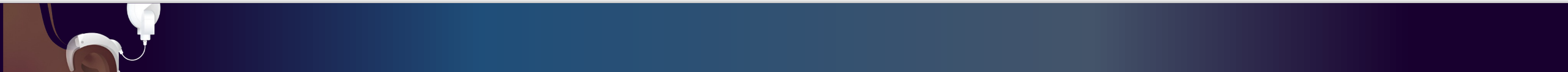
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Introduction



Variability in performances

Speech in noise, sound quality, quality of life..

Poor & good users

10-50% are classified as poor users (Moberly et al., 2016)

New requests from users

Sound quality, music perception

Individual factors

age at onset of hearing loss, aetiology, duration of moderate and severe-to-profound hearing loss…

Activate these factors?

Study **individual factors** that can be **manipulated** even for CI users already implanted

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Frequency-to-place factors

FTPM, Current Spread and Neural Responsiveness

Identify how these factors can be related to **requests** of CI users

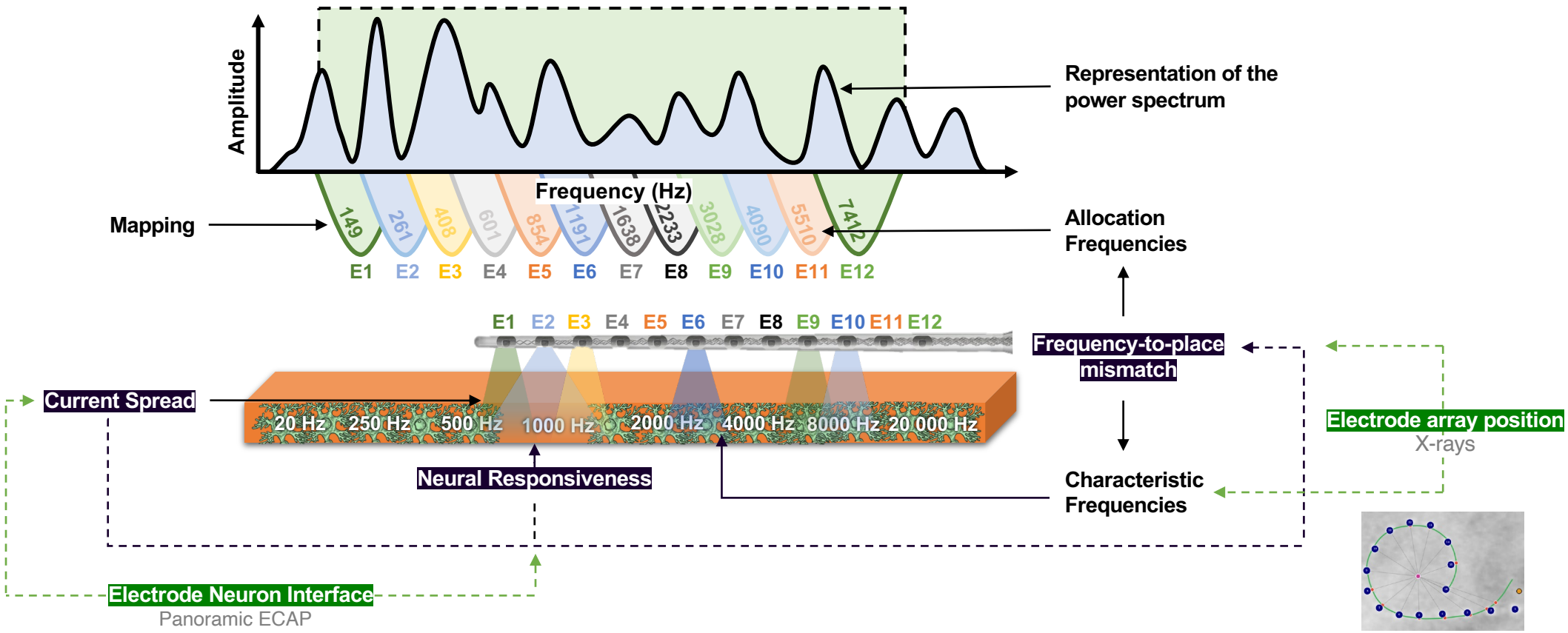
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Sound quality

Preference, pleasantness, Clarity..

Frequency-to-place factors

= factors related to how well information is distributed/conveyed to the auditory system



FTPM & Sound quality

JASA ARTICLE

Effect of frequency-to-place mismatch on speech and music sound quality in acoustic cochlear implant simulation

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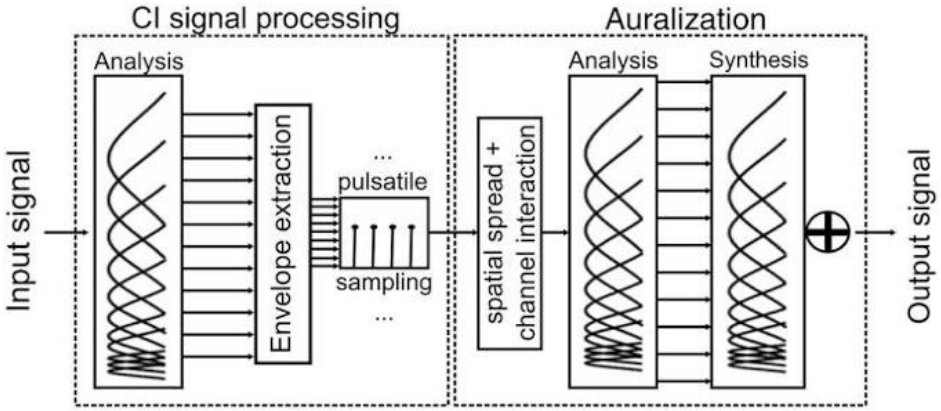
[3]*OticonA/S, Smørum, Denmark*

FTPM & Sound quality



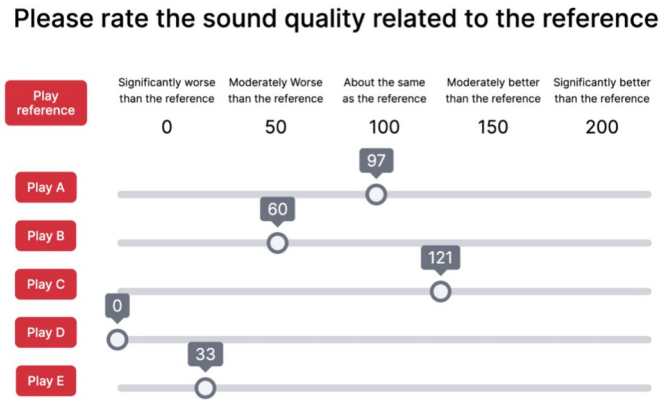
Music & Speech Vocoded

12 sounds in total



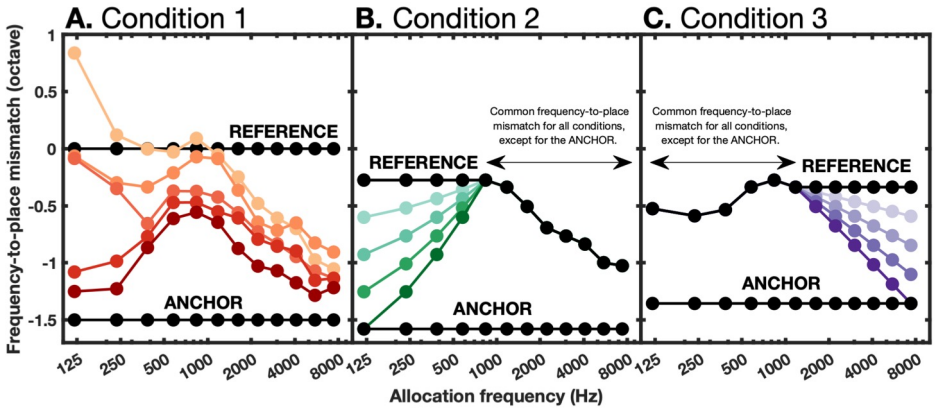
23 NH tested with a noise vocoder

Online and Onsite



MUSHRA used

1 MUSHRA per condition



Different conditions simulated

Realistic FTPM + Low and high Freq only

FTPM & Sound quality

Main results

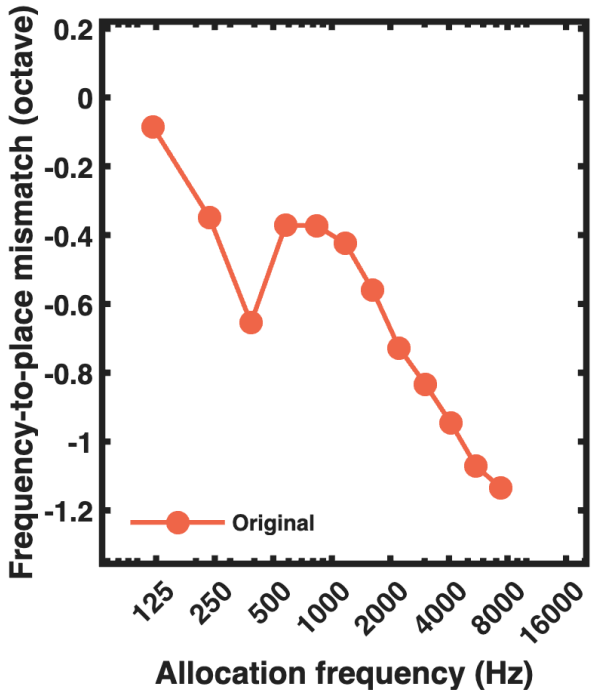
More mismatch = more sound quality degradation
No matter the sound

Mismatch has more effect in low frequencies
Compared to the high frequency condition

No effect of the interaction
FTPM / Sound Quality / Sound Features
We cannot predict the effect of the environment on FTPM/sound quality

FTPM variability has an effect on sound quality
Variability between adjacent electrodes degrades sound quality

Example of one FTPM profile



FTPM & Sound quality

Main results

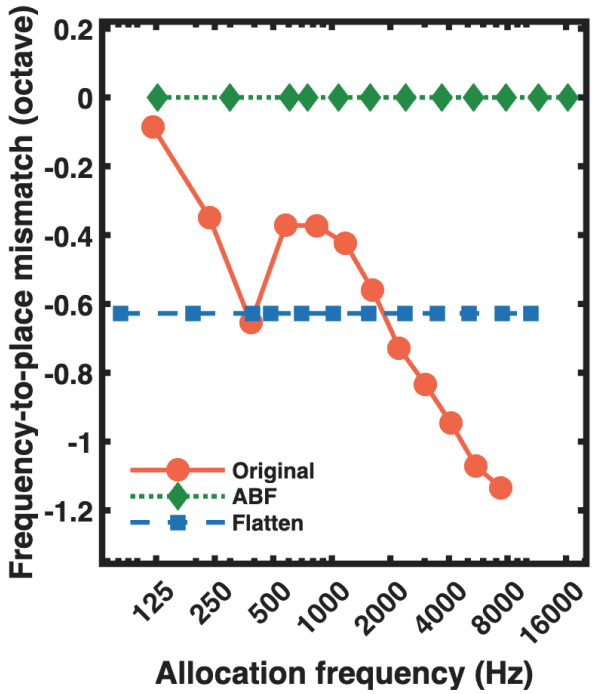
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Example of one FTPM profile



Current Strategy
Anatomy based frequency (ABF) mapping
Improve music perception but can remove some low-frequency information

Potential Alternative Strategy
"Flatten" the mismatch mapping
Improve music perception but can remove some low-frequency information



- Implanted 5 years ago
- Good in speech understanding
- **But still difficulties with sound quality**

How can we demonstrate/test the effect of these factors?

- **Realistic signal** Related to daily difficulties of CI users
- **Electrode level** We want one value per electrode!
- **Clinically adapted** Easy to understand and quick !!

↓ ↓

Test 1 ***Test 2***

Speech related test **Music related test**



- ❑ **Realistic signal** Related to daily difficulties of CI users
- ❑ **Electrode level** We need to know where is the problem
- ❑ **Clinically adapted** Easy to understand and quick !!

↓

Test 1

Speech related test

↓

Test 2

Music related test

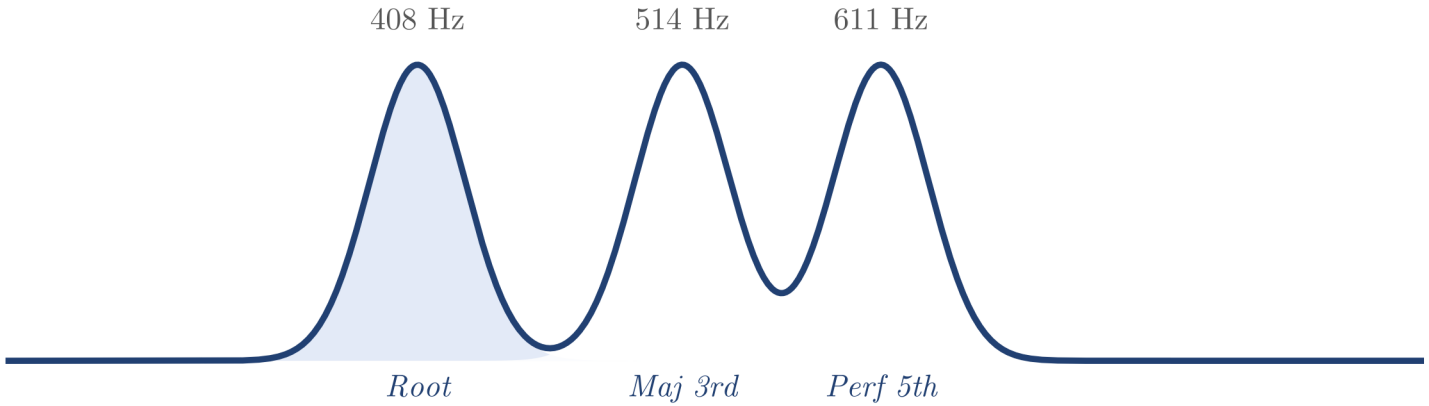


Signal used - Chords

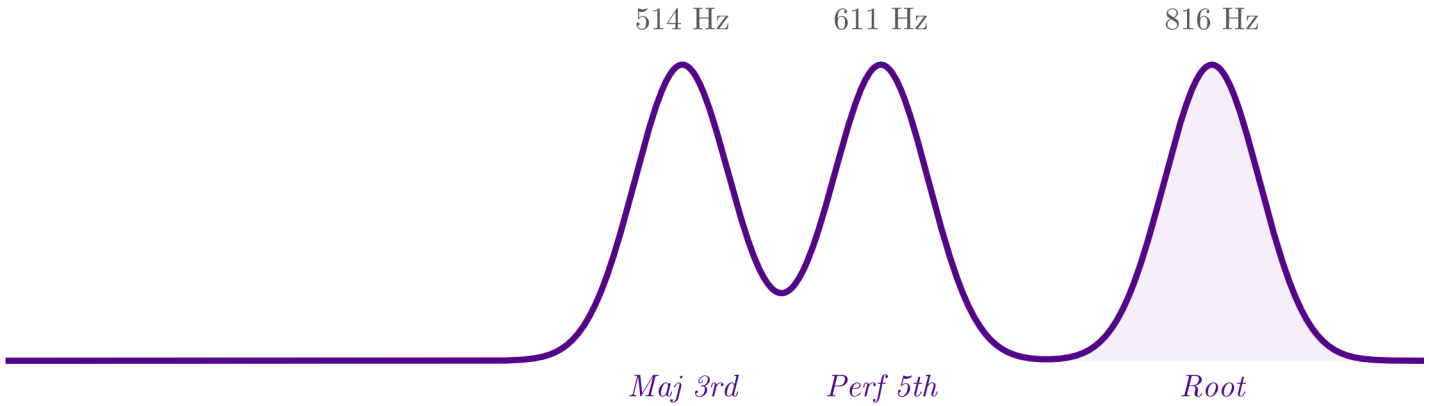


Inverted chords

Root position



First inversion

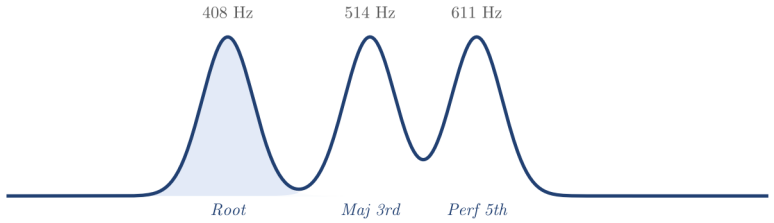


Chord 1	Chord 2

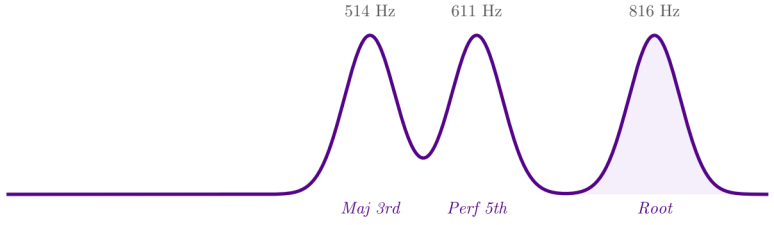
+5	+4	+3	+2	+1	=	+1	+2	+3	+4	+5

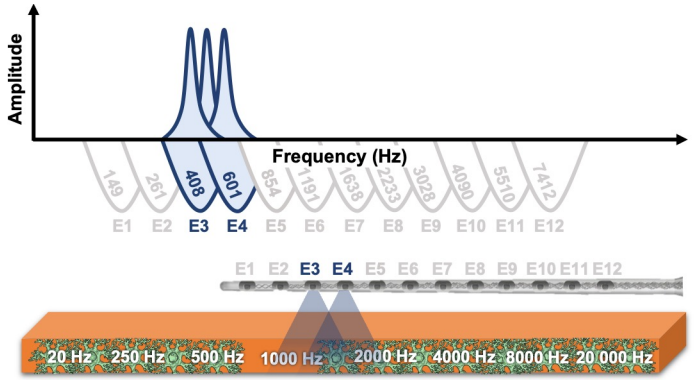
Which one of the two chords **is more pleasant?**

Root position



First inversion





From **paired comparisons** to **FIFs**

Paired-comparison between inverted chords

↓

Paired-comparison between "inverted" electrodes

↓

But we want one value of quality per electrode!

Elo rating system

51 languages

Article Talk Read Edit View history

From Wikipedia, the free encyclopedia

The **Elo**[a] **rating system** is a method for calculating the relative skill levels of players, originally designed for rating chess players. It is a special case of the Bradley–Terry model. The Elo system was invented as an improved chess rating system over the previously used Harkness rating system[1] and has since been adapted for use in other zero-sum games and sports, including tennis, association football (soccer), American football, baseball, basketball, pool, various board games and esports. It is named after its creator Arpad Elo, a Hungarian-American chess master and physics professor.



Arpad Elo, the inventor of the Elo rating system

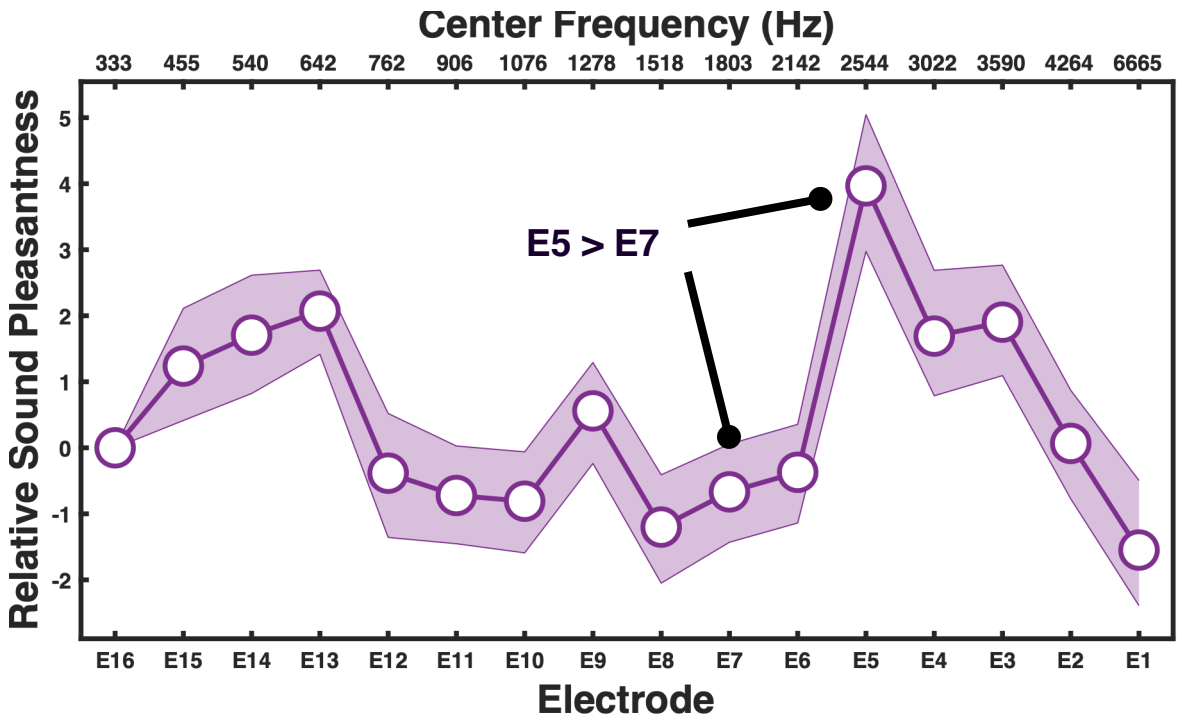
The difference in the ratings between two players serves as a predictor of the outcome of a match. Two players with equal ratings who play against each other are expected to score an equal number of wins. A player whose rating is 100 points greater than their opponent's is expected to score 64%; if the difference is 200 points, then the expected score for the stronger player is 76%.[2]

A player's Elo rating is a number that may change depending on the outcome of rated games played. After every game, the winning player takes points from the losing one. The difference



Bradley-Terry Luce algo - Very efficient method to optimize your test!

From **paired comparisons** to **FIFs**



Frequency-importance function (FIF)

Paired-comparison

using the 11 points bipolar rating

↓

Electrode comparison

Use the 11 points bipolar rating as **a proxy of electrode comparisons**

↓

Bayesian modelling

Probabilistic model able to classify objects (here electrodes) between each others, without the necessity to test all combinations

$$P(\text{prefer } i \text{ over } j) = \frac{\pi_i}{\pi_i + \pi_j}$$

↓

Electrode weight pleasantness for each electrode

For example, E5 has the best pleasantness

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Comparison with factors

FTPM, CS and NR

Participants

24 experienced CI users

10 “Pre-ECI”

Implanted **before language acquisition**

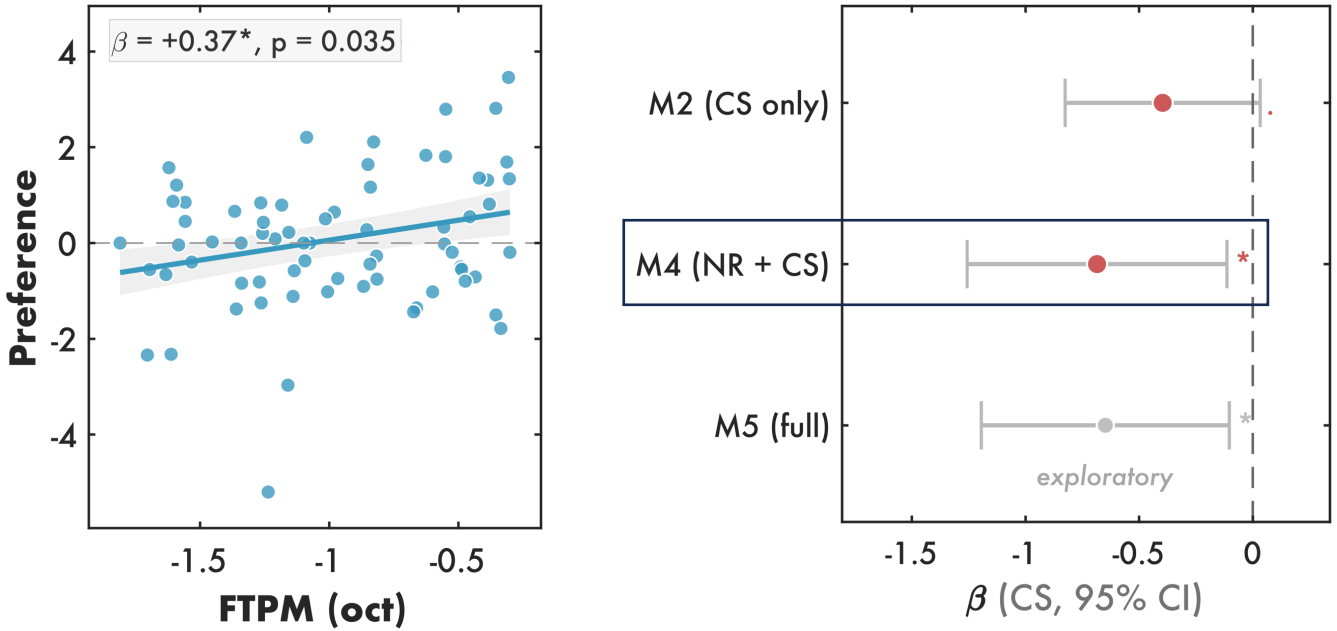
14 “Post-LCI”

Implanted **after language acquisition**



Perception of Chords

Only for Post-LCI



→ FTPM and CS significant **for Post-LCI**

more FTPM and CS decrease preference

→ Intervention on mapping?

For electrode with high absolute mismatch?

Case study

Can we really talk about lack of adaptation?



Case study

Adaptation process

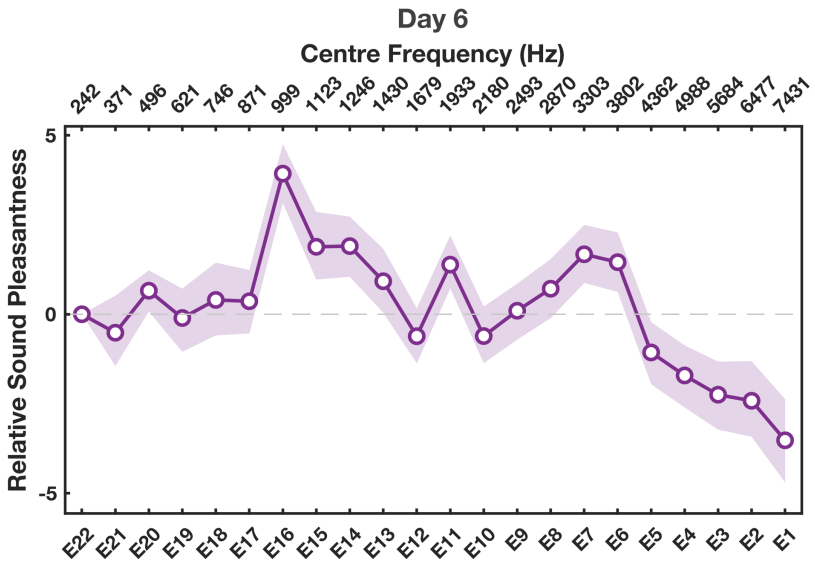
Case study on a newly implanted

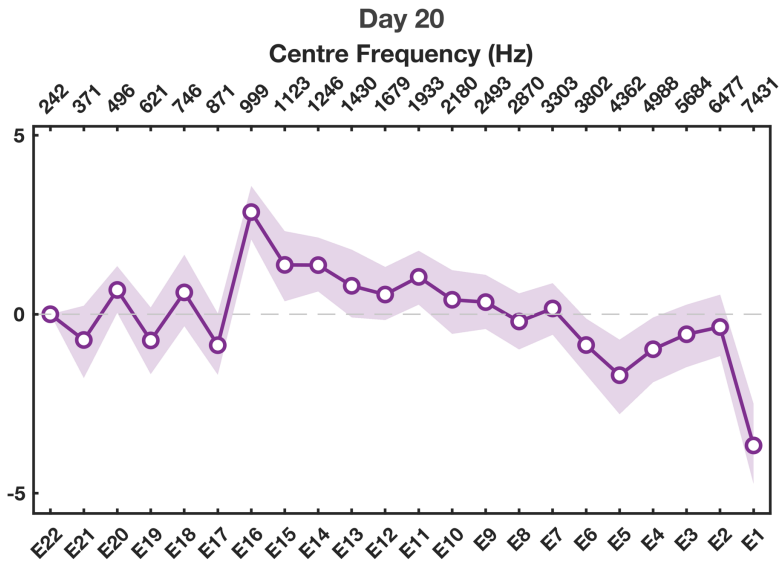
Sex	Female
Age at activation	26 years
Age at onset of deafness	11 years
Duration of deafness	15 years
Aetiology	Unknown
Device manufacturer and model	Cochlear
Number of active electrodes	22
Implanted ear	Left
Contralateral ear status	Hearing Aid

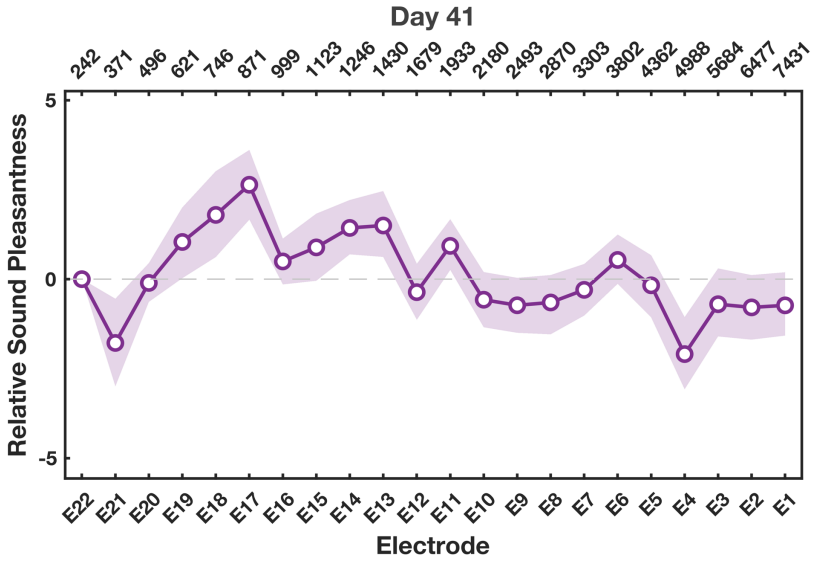


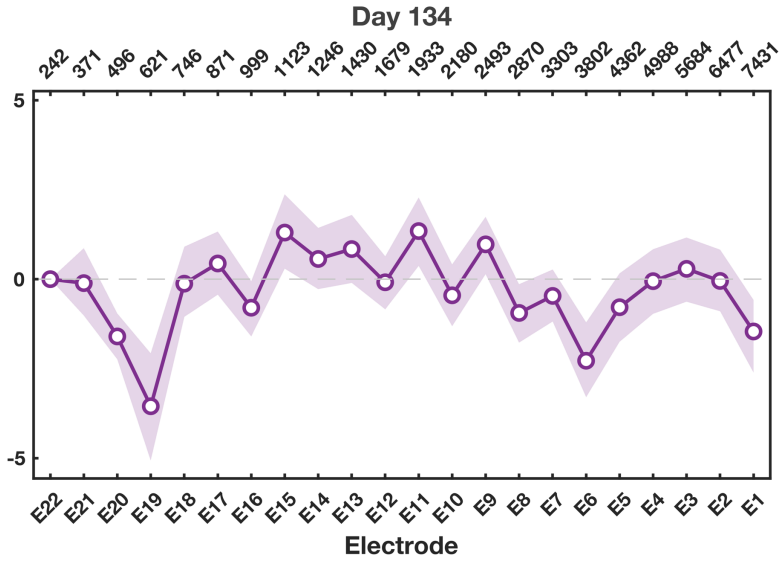
post switch-on

Case study

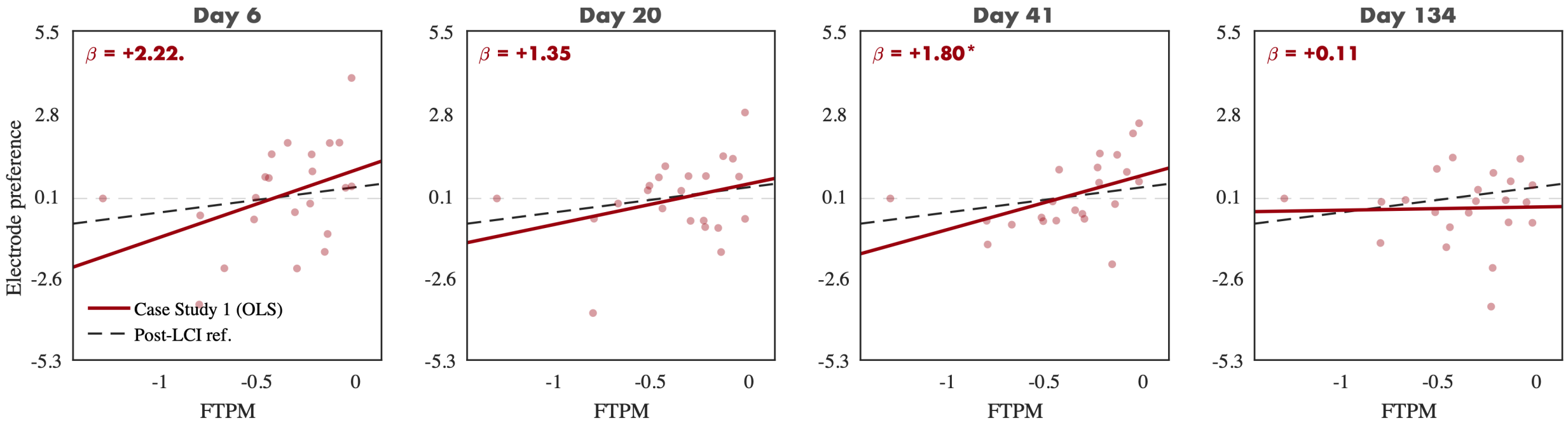

Day 6
Centre Frequency (Hz)
242 371 496 621 746 871 999 1123 1246 1430 1679 1933 2180 2493 2870 3303 3802 4362 4988 5684 6477 7431
Relative Sound Pleasantness
5
0
-5
E22 E21 E20 E19 E18 E17 E16 E15 E14 E13 E12 E11 E10 E9 E8 E7 E6 E5 E4 E3 E2 E1


Day 20
Centre Frequency (Hz)
242 371 496 621 746 871 999 1123 1246 1430 1679 1933 2180 2493 2870 3303 3802 4362 4988 5684 6477 7431
5
0
-5
E22 E21 E20 E19 E18 E17 E16 E15 E14 E13 E12 E11 E10 E9 E8 E7 E6 E5 E4 E3 E2 E1


Day 41
242 371 496 621 746 871 999 1123 1246 1430 1679 1933 2180 2493 2870 3303 3802 4362 4988 5684 6477 7431
Relative Sound Pleasantness
5
0
-5
E22 E21 E20 E19 E18 E17 E16 E15 E14 E13 E12 E11 E10 E9 E8 E7 E6 E5 E4 E3 E2 E1
Electrode


Day 134
242 371 496 621 746 871 999 1123 1246 1430 1679 1933 2180 2493 2870 3303 3802 4362 4988 5684 6477 7431
5
0
-5
E22 E21 E20 E19 E18 E17 E16 E15 E14 E13 E12 E11 E10 E9 E8 E7 E6 E5 E4 E3 E2 E1
Electrode

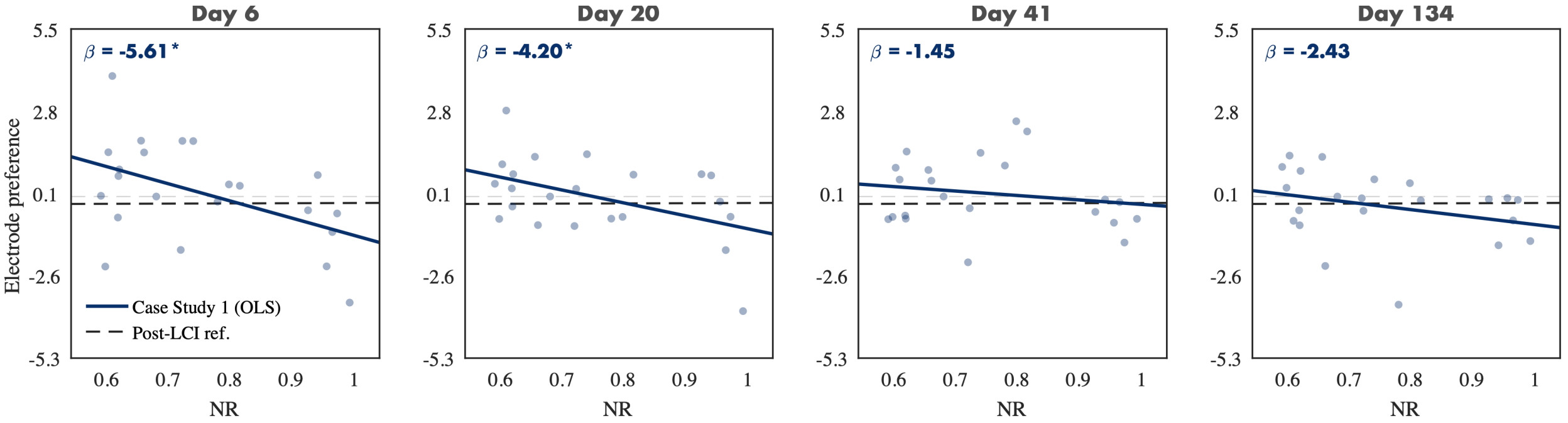
Case study



Electrodes associated with more absolute FTPM are getting more preferred over time

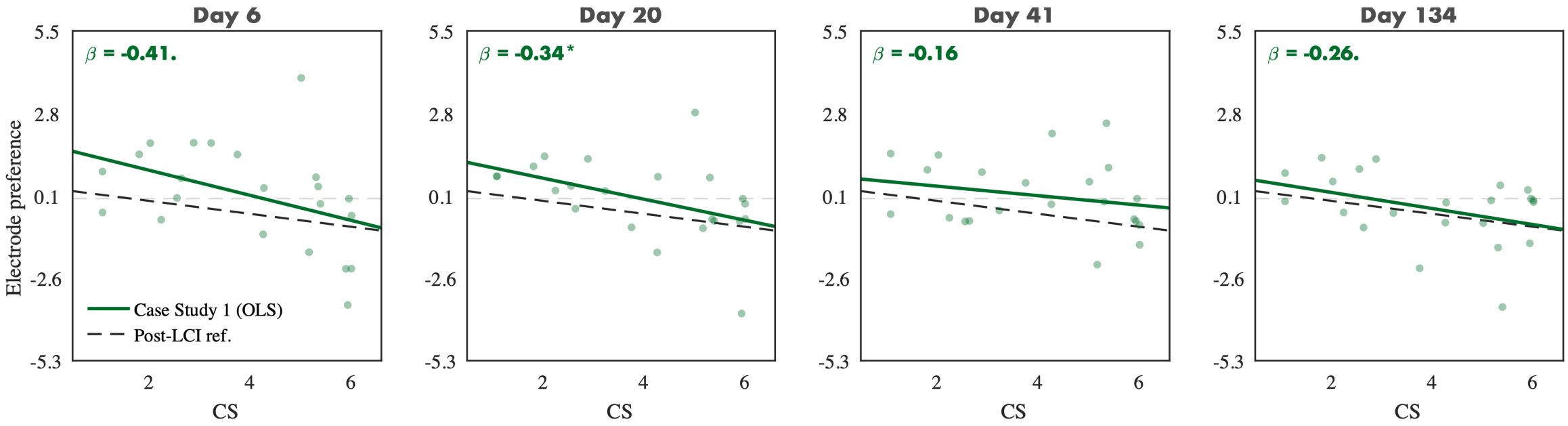
= proof of adaptation at the electrode level *(one case study)*

Case study



Electrodes with high NR *(good ENI)* are initially less preferred
this trend disappeared between day 20 and day 41

Case study



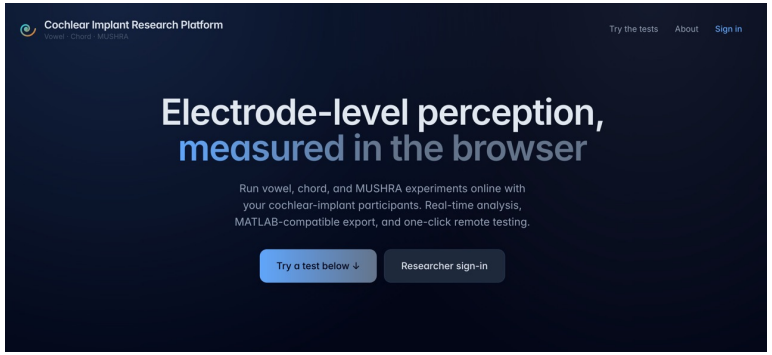
No real change over time for CS

Take away

→ Investigate **electrode factors** at the **electrode level**

→ Use **Bradley-Terry Luce method** if you want to optimize your test

→ We will deploy a **testing website** (chord + speech test)



https://very soon website.cam.ac.uk/

Thanks for your attention



SoundLab at IHI 2026

Research project funded by William Demant | Fonden